

# 3-Phase Gate Driver, 60 V

## NCD83591

### Overview

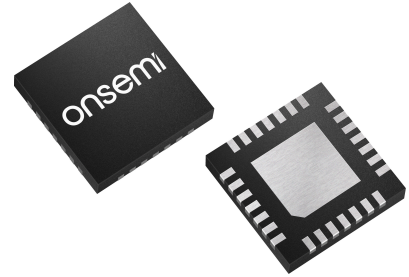
The NCD83591 is an easy to use, multi-purpose 3-phase gate-driver optimized for low BOM cost. The wide operating voltage range make this device ideal for industrial and commercial applications from 5 V (min) up to 60 V (max).

### Features

- Operation from 5 V to 60 V Supply Voltage with Tolerance from 4 V to 70 V
- Constant Current Drive of Power FET, Configurable from 5 mA to 250 mA (16 Settings by Input Level to VDRV Pin)
- Internally Generated Gate Drive Supply Voltages
- Tolerant to -12 V DC for High Side Gate and Source Output Pins
- Gate Sensing for Cross Conduction Protection and Optimized Dead Time
- Embedded High GBW General Purpose Amplifier (Intended for Current Sensing and Externally Configurable)
- Embedded Charge Pump Function Allowing DC Static Drive
- Strong Hold-off Current Preventing Self-turn-on
- Internal Gate Pull-down to GND during Loss of VDD (Core) Power
- Under Voltage Lock Out on VM, VGL, and VGH-VM with Optional Trip Levels
- 5 V/3.3 V Compatible 6 Input Control Plus Enable
- Individual Six Gate Control Mode with Separate Pin to Override and Drive All 6 Gate Outputs Low
- 4 kV HBM and 1 kV CDM ESD Protection
- This is a Pb-Free Device

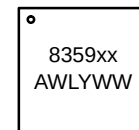
### Applications

- Power Tools
- Industrial Actuators and Cobots
- Automated Guided Robots
- Light Mobility (E-Bikes, E-scooters, Hoverboards)



QFN28 4x4, 0.4P  
CASE 485GF

### MARKING DIAGRAM



8359xx = Specific Device Code  
A = Assembly Location  
WL = Wafer Lot  
Y = Year  
WW = Work Week

### ORDERING INFORMATION

See detailed ordering and shipping information on page 9 of this data sheet.

# NCD83591

## APPLICATION CIRCUIT

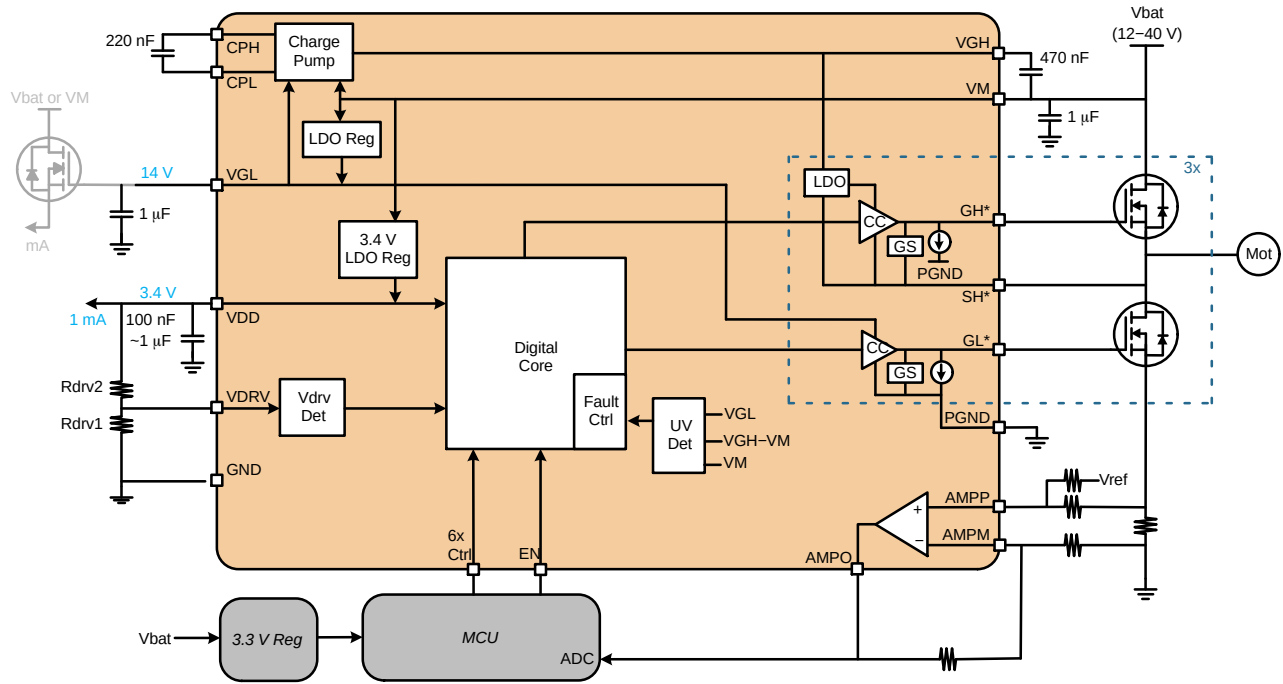


Figure 1. Typical Application Diagram

## PIN ASSIGNMENT

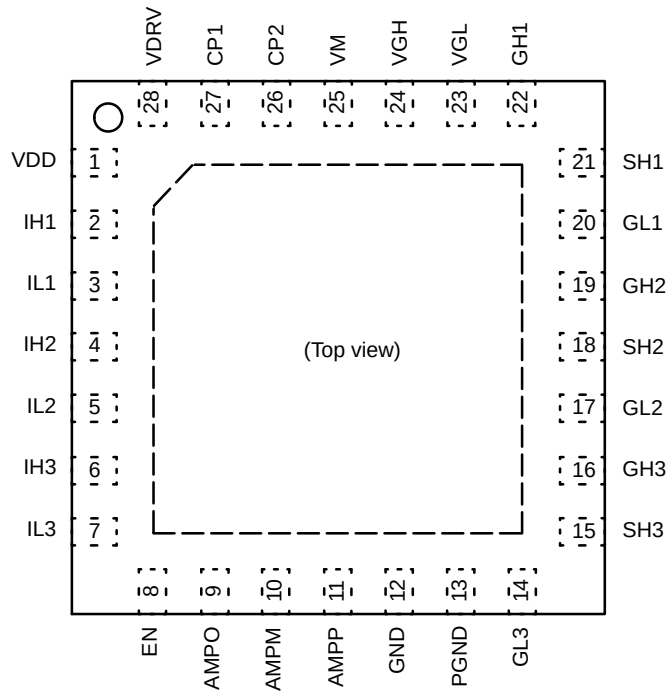


Figure 2. NCD83591 Pinout

**PIN ASSIGNMENTS & PIN DESCRIPTION**

| Pin # | Name | Description                                           |
|-------|------|-------------------------------------------------------|
| 1     | VDD  | Internal 3 V regulator bypass pin                     |
| 2     | IH1  | Active high control for GH1                           |
| 3     | IL1  | Active low control for GL1                            |
| 4     | IH2  | Active high control for GH2                           |
| 5     | IL2  | Active low control for GL2                            |
| 6     | IH3  | Active high control for GH3                           |
| 7     | IL3  | Active low control for GL3                            |
| 8     | EN   | Active high control to enable driver outputs          |
| 9     | AMPO | Configurable AMP output                               |
| 10    | AMPM | Configurable AMP negative input                       |
| 11    | AMPP | Configurable AMP positive input                       |
| 12    | GND  | General ground                                        |
| 13    | PGND | Power stage ground                                    |
| 14    | GL3  | Gate driver output for phase 3 low side external FET  |
| 15    | SH3  | Connection for motor phase 3                          |
| 16    | GH3  | Gate driver output for phase 3 high side external FET |
| 17    | GL2  | Gate driver output for phase 2 low side external FET  |
| 18    | SH2  | Connection for motor phase 2                          |
| 19    | GH2  | Gate driver output for phase 2 high side external FET |
| 20    | GL1  | Gate driver output for phase 1 low side external FET  |
| 21    | SH1  | Connection for motor phase 1                          |
| 22    | GH1  | Gate driver output for phase 1 high side external FET |
| 23    | VGL  | Internal 14 V regulator bypass pin                    |
| 24    | VGH  | Output of charge pump and storage capacitor pin       |
| 25    | VM   | Motor voltage                                         |
| 26    | CP2  | Pumping node of charge pump                           |
| 27    | CP1  | Switching node of charge pump                         |
| 28    | VDRV | Voltage used to control driver currents               |

**ABSOLUTE MAXIMUM RATINGS** (Voltages are referenced to GND unless otherwise noted)

| Parameter                                        | Pins                       | Ratings           | Unit |
|--------------------------------------------------|----------------------------|-------------------|------|
| Motor Supply Voltage                             | VM                         | -0.3 to 70        | V    |
| Gate Low Voltage                                 | VGL                        | -0.3 to 20        | V    |
| Gate High Voltage                                | VGH                        | -0.3 to 90        | V    |
| Gate High Voltage Relative to VM                 | VGH – VM                   | -0.3 to 20        | V    |
| High Side Output                                 | GH[1–3]                    | -12 to VGH + 0.3  | V    |
| Voltage between High Side Output and Motor Phase | GH[1–3] – SH[1–3]          | -0.3 to 20        | V    |
| Motor Phase                                      | SH[1–3]                    | -12 to VGH + 0.3  | V    |
| Low-side Output                                  | GL[1–3] (relative to PGND) | -0.3 to VGL + 0.3 | V    |
| Logic Power Supply                               | VDD                        | -0.3 to 3.65      | V    |
| Digital Inputs                                   | IH[1–3], IL[1–3], EN       | -0.3 to 6         | V    |
| Analog Inputs/Outputs                            | AMPM, AMPP, AMPO, VDRV     | -0.3 to VDD + 0.3 | V    |

**ABSOLUTE MAXIMUM RATINGS** (Voltages are referenced to GND unless otherwise noted) (continued)

| Parameter                                     | Pins | Ratings           | Unit |
|-----------------------------------------------|------|-------------------|------|
| Charge Pump – Switching Node                  | CP1  | –0.3 to VGL + 0.3 | V    |
| Charge Pump – Pumping Node                    | CP2  | VM – 1 to VGH + 1 | V    |
| Charge Pump – Pumping Node Peak Input Current | CP2  | –0.1 to 0.1       | A    |
| Storage Temperature                           |      | –55 to 150        | °C   |
| Junction Temperature                          |      | –40 to 150        | °C   |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

**ESD RATINGS**

| Parameter                  | Ratings | Unit |
|----------------------------|---------|------|
| Human Body Model (HBM)     | ±4000   | V    |
| Charged Device Model (CDM) | ±1000   | V    |

**THERMAL INFORMATION**

| Parameter | Pins                                         | Ratings | Unit |
|-----------|----------------------------------------------|---------|------|
| Theta-JA  | Junction Ambient Thermal Resistance          | 62.5    | °C/W |
| Psi-JB    | Junction to Board Characterization Parameter | 23.0    | °C/W |
| Psi-JT    | Junction to Top Characterization Parameter   | 4.2     | °C/W |

**ELECTRICAL CHARACTERISTICS**

(Valid at a junction temperature range from –40°C to 150°C, for VM supply Voltage 8.0 V ≤ VM ≤ 60 V unless otherwise specified. Typical values at 25°C and VM = 24 V unless specified otherwise.)

| Parameter               | Symbol | Conditions                                                                           | Min | Typ | Max            | Unit |
|-------------------------|--------|--------------------------------------------------------------------------------------|-----|-----|----------------|------|
| VM Supply Voltage Range |        | Normal mode                                                                          | 5   | 24  | 60<br>(Note 1) | V    |
|                         |        | Full logic functionality, driver stage off                                           | 4   | –   | 70             | V    |
| VM Supply Current       |        | EN = H (outputs not toggling),<br>AMP used                                           | –   | 4.6 | 5.7            | mA   |
| VM Supply Current       |        | EN = H (outputs not toggling), AMP<br>disabled by tying AMPP = VDD and<br>AMPM = GND | –   | 4.1 | 5.2            | mA   |

**INTERNAL REGULATOR (VDD)**

|                           |  |  |     |     |     |    |
|---------------------------|--|--|-----|-----|-----|----|
| VDD Output Voltage        |  |  | 3.2 | 3.4 | 3.6 | V  |
| VDD External Load Current |  |  | –   | –   | 1   | mA |

**GATE VOLTAGES (CP1, VGH, VGL )**

|                            |  |                                     |           |            |    |     |
|----------------------------|--|-------------------------------------|-----------|------------|----|-----|
| VGL Output Voltage         |  | I <sub>VGL</sub> < 15 mA, VM > 15 V | 13        | 14         | 15 | V   |
| VGL Output Voltage         |  | I <sub>VGL</sub> < 15 mA, VM < 15 V | VM – 0.43 | –          | –  | V   |
| VGL Current Limit          |  |                                     | –         | 23         | 30 | mA  |
| Charge Pump Frequency      |  |                                     | –         | 125        | –  | kHz |
| Charge Pump Output Voltage |  | VGH – VM, I <sub>VGH</sub> < 7.5 mA | VGL – 1.2 | VGL – 0.75 | –  | V   |

**GATE DRIVERS (GH[1–3], GL[1–3])**

|                                           |      |           |   |          |   |    |
|-------------------------------------------|------|-----------|---|----------|---|----|
| GL[n], GH[n] Discharge<br>Current to PGND | IDIS | “L” level | – | 2 x ICHG | – | mA |
|-------------------------------------------|------|-----------|---|----------|---|----|

**ELECTRICAL CHARACTERISTICS** (continued)

(Valid at a junction temperature range from -40°C to 150°C, for VM supply Voltage 8.0 V ≤ VM ≤ 60 V unless otherwise specified. Typical values at 25°C and VM = 24 V unless specified otherwise.)

| Parameter                              | Symbol            | Conditions                                                      | Min | Typ | Max  | Unit |
|----------------------------------------|-------------------|-----------------------------------------------------------------|-----|-----|------|------|
| <b>GATE DRIVERS (GH[1-3], GL[1-3])</b> |                   |                                                                 |     |     |      |      |
| GL[n], GH[n] Charge Current            | ICHG              | "H" level, VDRVcode = 15                                        | –   | 5   | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 14                                        | –   | 10  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 13                                        | –   | 15  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 12                                        | –   | 20  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 11                                        | –   | 25  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 10                                        | –   | 30  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 9                                         | –   | 40  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 8                                         | –   | 50  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 7                                         | –   | 60  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 6                                         | –   | 75  | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 5                                         | –   | 100 | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 4                                         | –   | 125 | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 3                                         | –   | 150 | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 2                                         | –   | 175 | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 1                                         | –   | 200 | –    | mA   |
|                                        |                   | "H" level, VDRVcode = 0                                         | –   | 250 | –    | mA   |
| Gate Sense Low Threshold               | VGSL              | GL[1-3], GH[1-3] – SH[1-3]                                      | –   | 1   | –    | V    |
|                                        |                   | Hysteresis                                                      | –   | 1   | –    | V    |
| Cross Conduction Protection Time       | t <sub>xcp</sub>  | See Figure 3<br>No Load on GL[1-3] or GH[1-3]                   | –   | 200 | –    | ns   |
| Input to Output Propagation Delay      | t <sub>prop</sub> | See Figure 3<br>No Load on GL[1-3] or GH[1-3]                   | –   | –   | 100  | ns   |
| Pull-down Current (GH*, GL*)           | IPD               | VM = 5 V<br>VDD = 0 V<br>GH* = GL* = 2 V<br>Applied per Table 1 | 500 | 900 | 1300 | μA   |

**CONFIGURABLE AMP (AMPP, AMPM, AMPO)**

|                           |          |                                         |      |                   |            |      |
|---------------------------|----------|-----------------------------------------|------|-------------------|------------|------|
| AMPP, AMPM Input Current  | AMPP_CUR | INPUT = 0–VDD                           | –1   | –                 | 1          | μA   |
|                           | AMPM_CUR | INPUT = 0–VDD                           | –1   | –                 | 1          | μA   |
| AMPO Full Scale Range_1   | FS1      | RL = 10 kΩ                              | 0.1  | –                 | VDD – 0.15 | V    |
| UGBW_1                    |          | RL = 10 kΩ to GND, CL = 60 pF<br>At FS1 | 10   | –                 | –          | MHz  |
| Open Loop Gain_1          |          | At FS1                                  | 90   | –                 | –          | dB   |
| AMPO Full Scale Range_2   | FS2      | RL = 10 kΩ                              | 0.05 | –                 | VDD – 0.1  | V    |
| UGBW_2                    |          | RL = 10 kΩ to GND, CL = 60 pF<br>At FS2 | 8    | –                 | –          | MHz  |
| Open Loop Gain_2          |          | At FS2                                  | 75   | –                 | –          | dB   |
| Slew Rate                 |          | Unity Gain, RL = 10 kΩ, CL = 60 pF      | 10   | –                 | –          | V/μs |
| Input Offset Voltage      |          | RL = 10 kΩ                              | –    | ±0.55<br>(Note 3) | ±4.0       | mV   |
| Common Mode Input Voltage |          |                                         | 0    | –                 | VDD – 1.2  | V    |

# NCD83591

## ELECTRICAL CHARACTERISTICS (continued)

(Valid at a junction temperature range from  $-40^{\circ}\text{C}$  to  $150^{\circ}\text{C}$ , for VM supply Voltage  $8.0\text{ V} \leq \text{VM} \leq 60\text{ V}$  unless otherwise specified. Typical values at  $25^{\circ}\text{C}$  and VM = 24 V unless specified otherwise.)

| Parameter | Symbol | Conditions | Min | Typ | Max | Unit |
|-----------|--------|------------|-----|-----|-----|------|
|-----------|--------|------------|-----|-----|-----|------|

### DIGITAL INPUTS (IH[1-3], IL[1-3], EN)

|                          |  |              |     |   |     |               |
|--------------------------|--|--------------|-----|---|-----|---------------|
| High-level Input Voltage |  | VIH (Note 2) | 2.1 | – | 5.5 | V             |
| Low-level Input Voltage  |  | VIL (Note 2) | 0   | – | 0.7 | V             |
| Pull-down Current        |  |              | 30  | – | 100 | $\mu\text{A}$ |

### PROTECTION LEVELS

|                                         |                   |                                                          |            |              |             |               |
|-----------------------------------------|-------------------|----------------------------------------------------------|------------|--------------|-------------|---------------|
| VM Under-voltage Lock Out               | VM <sub>UV</sub>  |                                                          | 4.3        | 4.5          | 4.7         | V             |
|                                         |                   | Hysteresis                                               | –          | 0.2          | –           | V             |
| VGL Under-voltage Lock Out              | VGL <sub>UV</sub> | VVGL Falling<br>NCD83591MNTXG<br>NCD83591MNATXG          | 3.9<br>7.7 | 4.1<br>8.0   | 4.3<br>8.3  | V             |
|                                         |                   | Hysteresis<br>NCD83591MNTXG<br>NCD83591MNATXG            | –<br>–     | 0.20<br>0.27 | –<br>–      | V             |
| VGH Under-voltage Lock Out              | VGH <sub>UV</sub> | VGH – VM Falling<br>NCD83591MNTXG<br>NCD83591MNATXG      | 2.8<br>7.4 | 3.0<br>8.0   | 3.25<br>8.4 | V             |
|                                         |                   | Hysteresis<br>NCD83591MNTXG<br>NCD83591MNATXG            | –<br>–     | 0.20<br>0.30 | –<br>–      | V             |
| Under-voltage Lock Out De-glitch Filter | UVFILT            | VM <sub>UV</sub> , VGL <sub>UV</sub> , VGH <sub>UV</sub> | –          | 200          | –           | $\mu\text{s}$ |
| VDD POR Lock Out                        | VDD <sub>UV</sub> |                                                          | 1.55       | 2.26         | 2.75        | V             |
|                                         |                   | Hysteresis                                               | –          | 0.15         | –           | V             |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Power dissipation and temp must be accounted for when connecting to voltages > 12 V.
2. Digital inputs are 5 V tolerant.
3. Typical value is a 1 sigma number.

## DETAILED FUNCTIONAL DESCRIPTION

## Chip Activation, System States and Shutdown

Table 1 shows the operation modes and associated pin states.

Table 1. OPERATION MODES

| State     | POR<br>(Note 4) | UV_VM<br>(Note 3) | UV_VGL<br>(Note 2, 3) | UV_VGH<br>(Note 3) | Digital Inputs |     |                 | Driver Outputs<br>(Note 2) |      | CP1<br>(Note 2) |
|-----------|-----------------|-------------------|-----------------------|--------------------|----------------|-----|-----------------|----------------------------|------|-----------------|
|           |                 |                   |                       |                    | EN             | IH* | IL*<br>(Note 1) | GH*                        | GL*  |                 |
| POR       | H               | X                 | X                     | X                  | X              | X   | X               | L-PD                       | L-PD | L               |
| UV        | L               | H                 | X                     | X                  | X              | X   | X               | L-PD                       | L-PD | L               |
| UV        | L               | X                 | H                     | X                  | X              | X   | X               | L-PD                       | L-PD | L               |
| Charge    | L               | L                 | L                     | H                  | X              | X   | X               | L-PD                       | L-PD | Active          |
| Disable   | L               | L                 | L                     | L                  | L              | X   | X               | L-PD                       | L-PD | Active          |
| Drive     | L               | L                 | L                     | L                  | H              | L   | L               | L                          | H    | Active          |
| Freewheel | L               | L                 | L                     | L                  | H              | L   | H               | L                          | L    | Active          |
| Freewheel | L               | L                 | L                     | L                  | H              | H   | L               | L                          | L    | Active          |
| Drive     | L               | L                 | L                     | L                  | H              | H   | H               | H                          | L    | Active          |

1. IL\* inputs are active low.
2. Active means outputs are following the behavior of normal operation. L-PD means that pull down currents (IPD) may be active if 3 V core supply is not sufficient to drive L.
3. H = Under Voltage condition (supply voltage is less than detection levels & filter timer has expired).
4. H = Reset state (VDD voltage is less than detection level).

## System Power Supplies (VM, VDD, VGL, VGH)

## Internal Core Regulator (VDD)

The internal regulator is supplied by VM and regulates to 3.4 V at VDD. VDD needs to be bypassed to GND for stability. It is not intended for any external loads (other than Rdrv1/Rdrv2), but can support a load up to 1 mA, if necessary.

## Gate Voltage Regulator (VGL)

The gate voltage regulator is supplied by VM and regulates to 14 V at VGL.

VGL provides the drive voltage for the low-side drivers GL[1-3] directly and for the high side drivers GH[1-3] through the charge pump circuitry. The output is current limited. The output at VGL must be bypassed with a capacitor to GND which should be at least 10 times the maximum gate capacitance of the power FETs.

## Charge Pump (VGH)

An on-chip charge pump provides the gate voltage for the high-side drivers.

## Motor Control Inputs (IL[1-3], IH[1-3], EN)

Once the NCD83591 is in Drive or Freewheel mode, a microcontroller can facilitate motor control via the inputs IL[1-3], IH[1-3] and EN. All pins are 5 V/3.3 V compatible. All pins have pull-down currents, so when left floating will be pulled low and cause the outputs to follow the appropriate row in Table 1.

## Gate Drive (GH[1-3], GL[1-3], SH[1-3])

The gate drive circuit of the NCD83591 includes 3 half-bridge drivers which control six external N-Channel FETs. The high side gate drivers GH[1-3] switch their gate connection either to VGH or the respective phase connection SH[1-3]. The low-side gate drivers GL[1-3] are switched from VGL to PGND.

## Gate Drive Source/Sink Current Selection

The gate drive currents are programmable to 16 different values as defined in the Electrical Characteristics Table. The VDRVcode is achieved by configuring Rdrv1 and Rdrv2 with standard 5% resistors as described in the Application Diagram with the values that match the desired gate drive currents from Table 2.

Table 2. VDRVcode DESCRIPTION

| VDRVcode | Rdrv1 | Rdrv2 | ICHG | IDIS |
|----------|-------|-------|------|------|
| 15       | 82k   | 12k   | 5    | 10   |
| 14       | 68k   | 15k   | 10   | 20   |
| 13       | 51k   | 16k   | 15   | 30   |
| 12       | 47k   | 20k   | 20   | 40   |
| 11       | 33k   | 18k   | 25   | 50   |
| 10       | 47k   | 33k   | 30   | 60   |
| 9        | 27k   | 24k   | 40   | 80   |
| 8        | 24k   | 27k   | 50   | 100  |
| 7        | 36k   | 51k   | 60   | 120  |
| 6        | 20k   | 36k   | 75   | 150  |
| 5        | 24k   | 56k   | 100  | 200  |
| 4        | 18k   | 56k   | 125  | 250  |
| 3        | 13k   | 56k   | 150  | 300  |
| 2        | 12k   | 82k   | 175  | 350  |
| 1        | 6.8k  | 91k   | 200  | 400  |
| 0        | 0     | Open  | 250  | 500  |

Selected Isrc currents will typically be between 1 mA/nC and 2 mA/nC of FET total gate charge (Qg) to achieve a power stage switching net (SH[1–3]) transition times of about 50 ns. General procedure to selecting a drive current is to start low and increase until ringing on the gate nets or switching nets becomes noticeable, but is still acceptable.

#### Gate Sensing and Cross Conduction Protection

The gate sensing and cross conduction protection features of the NCD83591 eliminate the need for any minimum dead time control by the gate driver, or even any dead times programmed into the MCU. Corresponding IH[n] and IL[n] pins can be pulsed at the same time and can even be shorted together for motor control methods that do not require tri-stating the power stage (drive 3 mode).

The cross conduction protection feature requires one side of a given phase to be proven to be in an “off” state before the other side is allowed to be turned on to prevent any cross conduction or shoot-through current in the power stage. The diagram in Figure 3 demonstrates this behavior.

In addition, if a given side is currently in an “off” state and something external pulls it to an “on” state, the part forces the opposite side to an “off” state to again prevent shoot-through currents. Sensing a FET is in an “off” state is done through the gate sensing circuit.

The MCU can still force a controlled dead time if desired. This is considered a Drive 6 configuration where each IL[n] and IH[n] pin is driven independently, and the MCU forces a dead time between when low and high sides are active. However, the cross conduction protection will still take precedence if this MCU programmed dead time becomes too short.

The gate sensing provides yet another feature of the part in which a much stronger sinking driver is activated about 50 ns after a gate is sensed to be low. This results in a low hold-off driver resistance to further prevent self turn-on when the SH[n] net is transitioning fast.

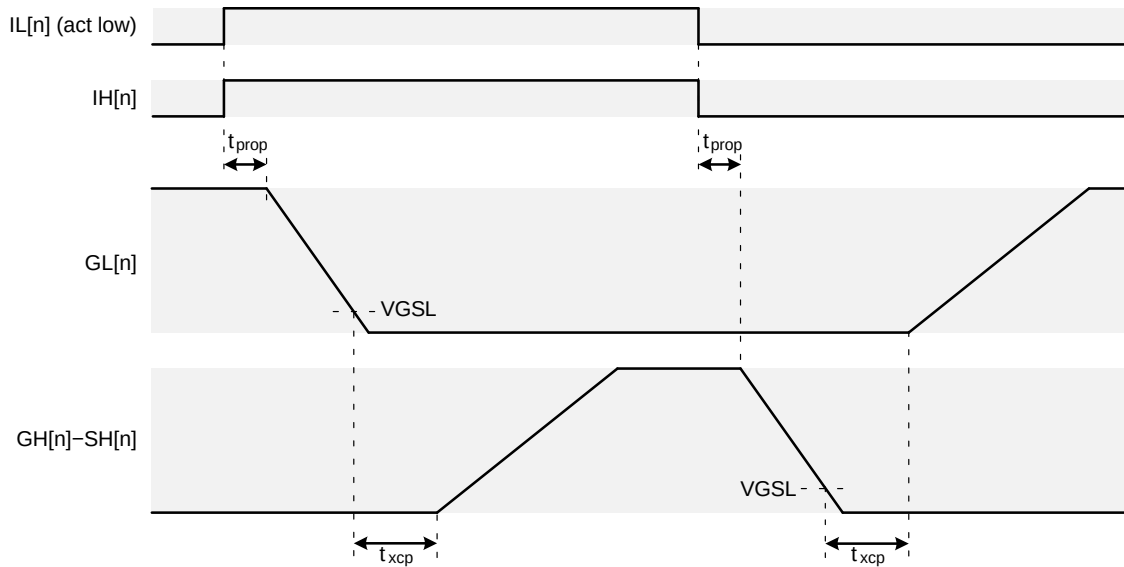


Figure 3. Gate Sensing and Dead Time Diagram

#### Configurable Amp (AMPP, AMPN, AMPO)

An externally configurable high performance general purpose amplifier is included. It is intended for current sensing as per the Application Diagram, but can be used in

any configuration that follows the constraints outlined in the Electrical Characteristics table.

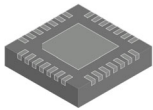
If the configurable amp is not intended to be used, about 0.5 mA of current can be saved by tying AMPP = VDD and AMPM = GND.

## NCD83591

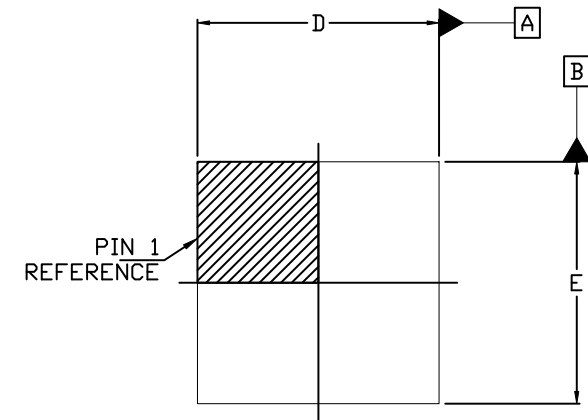
### ORDERING INFORMATION

| Device Order Number | Marking | UVOL                          | Package Type       | Shipping <sup>†</sup> |
|---------------------|---------|-------------------------------|--------------------|-----------------------|
| 1NCD83591MNTXG      | 8359A   | 4.1 V (VGL)<br>3.0 V (VGH-VM) | QFN28<br>(Pb-Free) | 4000 / Tape & Reel    |
| 1NCD83591MNATXG     | 8359AA  | 8.0 V                         | QFN28<br>(Pb-Free) | 4000 / Tape & Reel    |

<sup>†</sup>For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

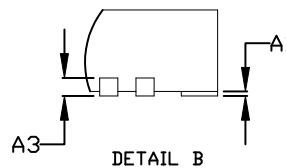

**QFN28 4x4, 0.4P**  
CASE 485GF  
ISSUE O

DATE 03 NOV 2020



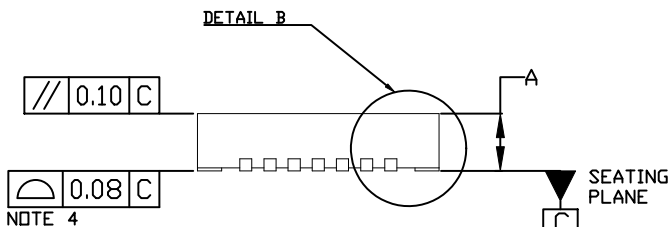
TOP VIEW

- NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2009.
  2. CONTROLLING DIMENSION: MILLIMETERS
  3. DIMENSION  $b$  APPLIES TO PLATED TERMINALS AND IS MEASURED BETWEEN 0.15 AND 0.30MM FROM THE TERMINAL TIP.
  4. COPLANARITY APPLIES TO THE TERMINALS.
  5. POSITIONAL TOLERANCE APPLIES TO ALL TERMINALS.
  6. PACKAGE DIMENSIONS EXCLUSIVE OF BURRS AND MOLD FLASH.

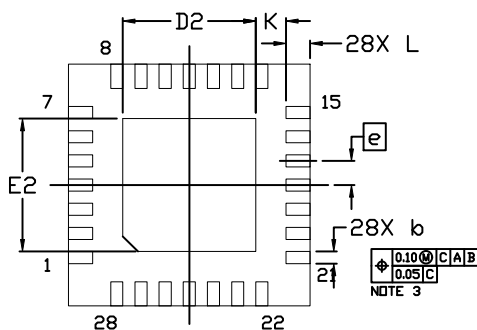


DETAIL B

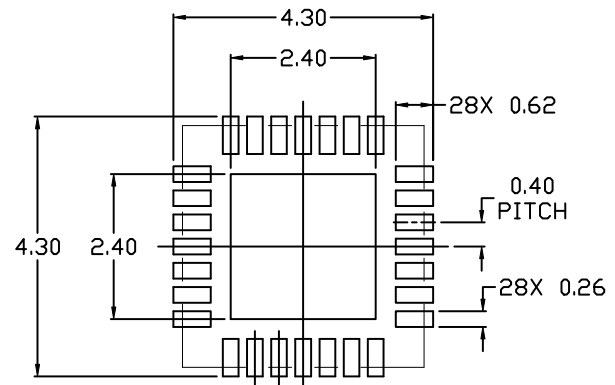
| DIM | MILLIMETERS |      |      |
|-----|-------------|------|------|
|     | MIN.        | NOM. | MAX. |
| A   | 0.80        | 0.90 | 1.00 |
| A1  | 0.00        | ---  | 0.05 |
| A3  | 0.20 REF    |      |      |
| b   | 0.15        | 0.20 | 0.25 |
| D   | 3.90        | 4.00 | 4.10 |
| D2  | 2.10        | 2.20 | 2.30 |
| E   | 3.90        | 4.00 | 4.10 |
| E2  | 2.10        | 2.20 | 2.30 |
| e   | 0.40 BSC    |      |      |
| K   | 0.50 REF    |      |      |
| L   | 0.30        | 0.40 | 0.50 |



SIDE VIEW

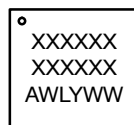


BOTTOM VIEW



RECOMMENDED MOUNTING FOOTPRINT\*

\* For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

**GENERIC  
MARKING DIAGRAM\***


XXXX = Specific Device Code  
A = Assembly Location  
WL = Wafer Lot  
Y = Year  
WW = Work Week

\*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "•", may or may not be present. Some products may not follow the Generic Marking.

|                         |                        |                                                                                                                                                                                  |
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| <b>DESCRIPTION:</b>     | <b>QFN28 4x4, 0.4P</b> | <b>PAGE 1 OF 1</b>                                                                                                                                                               |

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